MCQ solutions

1. **A.** By linearity, $E(Y) = 2E(X) - 1 = 5$. The constant is subtracted once.
2. **C.** The variance is multiplied by $(-3)^2 = 9$, and a translation does not change it. It cannot be negative.
3. **B.** The sum has variance $n\sigma^2$, then division by n multiplies the variance by $1/n^2$.
4. **D.** Linearity of expectation does not require independence. Additivity of variances requires additional conditions.
5. **C.** $E(X) = 1$, $E(X^2) = 2$, so $V(X) = 2 - 1^2 = 1$.
6. **A.** It is an upper bound on a deviation from the expectation, not an equality nor a bound applying directly on X.
7. **D.** Solve $25/n \leq 0.05$ with $n > 0$ gives $n \geq 500$. This is a sufficient size according to this bound.
8. **B.** A probability is always at most 1; the upper bound 2 gives no improvement.
9. **C.** This is convergence in probability, not a decrease of the error at each observa-
10. **A.** The mean is exactly X. Repeating the same variable does not create new independent observations.

True/false. False: the upper bound may be crude. True: take the square root of $a^2V(X)$. False: question 10, where the variance does not decrease.

Solutions to the applications and guidance

Application 1. Sum of probabilities 1. E=2; $E(X^2) = 11/2$, therefore $V = 3/2$ and $\sigma = \sqrt{6}/2$.

Application 2. $E(Y) = 4$, $V(Y) = 27/2$; $E(Z) = -2$, $V(Z) = 3/2$. The squared deviations from the mean do not change under a change of sign.

Application 3. Independence: variance 1 $/(4n)$. Identical copies: the mean equals X, so the variance 1/4 whatever n. Repeating the same measurement does not create an independent sample.

Application 4. The bound is 9/ $(0{,}25n) = 36/n$. Requiring $36/n \leq 0.01$ gives $n \geq 3600$.

Application 5. The event is $X \leq 2$ or $X \geq 8$. By symmetry, its probability is 2 $(1 + 10 + 45)/2^{10} = 7/64 \simeq 0{,}109375$. The bound equals $(1/40)/(9/100) = 5/18 \simeq 0{,}2778$, so it is correct but less precise.

Application 6. Theoretical standard deviations are about 0 ,1581, 0,05 and 0,01581. The simulated outputs fluctuate: we are comparing a set of means, not the successive values of a single mean.
Empirical proportions are not proofs. The reproducible results file
gives the values obtained with the chosen seed.

Application 7. The results 0, 1 give frequency 1 /2; adding 1 gives 2 /3, further from 1 /2. For every fixed precision, the probability of a deviation at least equal to this precision tends to 0 as n tends to infinity. This does not impose a monotonic approach on each realisation.

Reading the guarantees

Have students circle the strict inequality of the central event and the non-strict one of its complement. Always compare a computable probability with a universal bound. Do not present the simulation as a proof of the law of large numbers.

Teaching notes and solutions

Guidance for the teacher

Written test 5 and the second mock exam come before S73: do not ask for Bienaymé-Chebyshev or the law of large numbers. For the variance, have students circle the independence assumption. For the concentration, distinguish exact probability from upper bound, then a sufficient index from the actual minimal index. A simulation is not a proof, and an observed mean does not necessarily get closer with each new trial.

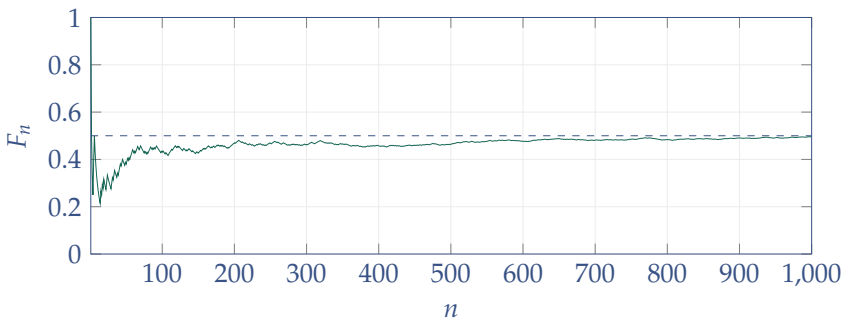
Solutions to the activity

1.

```
from random import Random

def trajectoire_piece(N=1000, graine=2026):
    rng = Random(graine)
    piles, frequences = 0, []
    for n in range (1, N+1):
        piles += rng.random() < 0.5
        frequences.append(piles/n)
    return frequences
```

Plot the points (n, F_n) with the list obtained. After two trials giving one head and one tail, $F_2 = \frac{1}{2}$. Whatever the third outcome, F_3 equals $\frac{1}{3}$ or $\frac{2}{3}$: the gap to $\frac{1}{2}$ increases. The stabilisation is not a monotonic decrease of the deviation along each path.



A simulated path, seed 2026; the line is $y = \frac{1}{2}$.

2. Denote by B_i the indicator of heads. Then $F_n = \frac{B_1+\cdots+B_n}{n}$. Each B_i has expectation $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$; by independence and change of scale, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.

3. Bienaymé-Chebyshev gives $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05) \leq \frac{1}{4\times 1000\times 0{,}05^2} = 0{,}1$. It is an upper bound, not the exact probability.

4. To ensure a complementary probability of at most 0 ,01, it suffices that $\frac{1}{4n\times 0{,}05^2} \leq 0{,}01$. This is equivalent to $n \geq 10000$. Then $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.

5. The simulation illustrates one finite realisation; other seeds give other trajectories. The inequality establishes a guarantee on the probability, for any sample satisfying the assumptions of independence and identical distribution.

E1. The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$; the shift -1 does not change the spread.

E2. By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4 - 5 = -1$. The independence of X, Y allows us to add the variances of $2X$ and $-Y$: $V(2X-Y) = 4V(X) + V(Y) = 4 + 4 = 8$.

E3. The mean of 64 independent identically distributed variables keeps expectation 10. Its variance is $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

E4. Bienaymé-Chebyshev with $\delta = 6$ gives $P(|X-100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound does not give an equality or an exact value.

E5. We have $P(|M_n - \mu| \geq 0{,}2) \leq \frac{4}{n \times 0{,}2^2} = \frac{100}{n}$. For this bound to be at most $0{,}05$, it suffices to have $n \geq 2000$. The complement then has probability at least 0.95.

Solutions to homework DM14

1. Denote by B_i the independent success indicators. $F_n = (\frac{1}{n}) \sum_{i=1}^{n} B_i$, with $E(B_i) = p$, $V(B_i) = p(1-p)$. Hence

$$E(F_n) = \frac{1}{n} \sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2} \sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$

2. $p(1-p) = p - p^2 = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$. Equality is attained for $p = \frac{1}{2}$.

3. For every $p \in [0,1]$, $P(|F_n - p| \geq 0{,}02) \leq \frac{1}{4n \times 0{,}02^2}$. It suffices that this bound is at most $0{,}05$, i.e. $n \geq 12500$. Taking the complement, this size guarantees the required probability, uniformly in p.

4. On $[0, 0{,}1]$, the function $p \mapsto p(1-p)$ is increasing because its derivative $1 - 2p$ is positive. So $p(1-p) \leq 0{,}1 \times 0{,}9 = 0{,}09$. It now suffices to have $n \geq \frac{0{,}09}{0{,}05 \times 0{,}02^2} = 4500$. These sizes are sufficient in the sense of the upper bound, not necessarily minimal for the exact probability.

5. If all trials copy the same variable B, then $F_n = \frac{B + \cdots + B}{n} = B$. Hence $V(F_n) = p(1-p)$, without division by n. Adding variances was allowed only because of independence; this assumption disappears here. For $0 < p < 1$, increasing the number of copies brings no new information.

Solutions to the textbook or past-paper work

M_N is the average number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6{,}63}{N}$.

$$P(|M_N - 7{,}14| \geq 0{,}14) \leq \frac{6{,}63}{0{,}14^2 N}.$$

To make this bound at most 0.05, it is enough to take

$$N \geq \left\lceil \frac{6{,}63}{0{,}05 \times 0{,}14^2} \right\rceil = 6766.$$

By complement, $P(7 < M_N < 7{,}28) \geq 0{,}95$.